

MongoDB CRUD & Analytics Dashboard Training Module

Hands-On Exercises: Import/Export, Projections, Upserts & Metabase Setup

AUDIENCE TARGET: ENGINEERING TEAM MEMBERS & STUDENTS

1. MongoDB CRUD & Data Import/Export Overview

This training module guides engineers through MongoDB CRUD operations, bulk import/export, filtering, projections, atomic updates, upserts, aggregation pipelines, and Metabase dashboard visualization using the public MongoDB MFlx dataset (~20,000 documents).

2. Data Import & Granular Export Commands

Exporting Only Certain Attributes:

```
mongoexport --db=mflix_db --collection=movies --type=csv --fields='title,year,imdb.rating' --out=movies_attr.csv
```

Exporting Only Certain Records (Query Filtering):

```
mongoexport --db=mflix_db --collection=movies --type=json --query='{ "genres": "Action", "year": { "$gt": 2015 } }' --out=action_2015.json
```

Exporting Certain Attributes of Certain Records:

```
mongoexport --db=mflix_db --collection=movies --type=json --query='{ "cast": "Leonardo DiCaprio" }' --fields='title,year,cast' --out=dicaprio.json
```

3. Insert, Update, Upsert & Delete Reference

Atomic Update (\$set, \$inc, \$push):

```
db.movies.updateOne( { title: 'Inception' }, { $set: { status: 'Blockbuster' }, $inc: { 'imdb.votes': 1 }, $push: { cast: 'New Actor' } } );
```

Upsert (Insert or Update):

```
db.users.updateOne( { email: 'user@example.com' }, { $set: { name: 'Alice', tier: 'VIP' }, $setOnInsert: { created_at: new Date() } }, { upsert: true } );
```

4. Open-Source Analytics Dashboards Comparison

Dashboard Tool	License	Key Strengths	Best Use Case
Metabase	AGPL / OS	No-code visual query builder, native MongoDB driver	Business Analytics & Interactive Dashboards

Redash	BSD / OS	Native MongoDB JSON query runner, SQL join support	Technical DBA Analytics & SQL Queries
Apache Superset	Apache 2.0	Enterprise BI visualization library	Multi-source enterprise reporting
Grafana	AGPL / OS	Real-time metrics & operational telemetry	System monitoring & cluster metrics